

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India — Expected Aug. 2026

CGPA: 8.19 / 10.0

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS, MATLAB, Bash

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, RAG Pipelines, Segment Anything (SAM)

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, GraphQL, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Git/GitHub, Linux, Firebase, Vector Databases (FAISS, DiskANN), Vercel

Research Experience

Undergraduate Researcher

June 2023 – Aug. 2023

Supervised by Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline to optimize **Segment Anything Model (SAM)** for agricultural crop-weed segmentation on sparse datasets, improving segmentation quality on novel agricultural imagery through targeted data preprocessing.
- Engineered custom image enhancement techniques including contrast adjustment, noise reduction, and edge detection to maximize SAM's zero-shot segmentation performance.
- Improved batch processing throughput by implementing vectorized NumPy operations and multiprocessing for large-scale image datasets (10,000+ images).

Industry Experience

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Collaborated with one other intern to architect **FastAPI** backend parsing 50,000+ lines of code daily across 15+ programming languages via Abstract Syntax Tree (AST) analysis, enabling automated real-time documentation generation across polyglot codebases.
- Designed hash-based change detection system using **SHA-256 on AST nodes**, chunking codebases function-by-function and skipping unchanged functions – reducing AI embedding generation costs by **85%** by eliminating redundant vector database updates.
- Built **RAG pipeline** using LangChain, Azure OpenAI embeddings, and Cosmos DB with DiskANN vector search for semantic code retrieval, migrating from FAISS as repository scale grew.
- Delivered 12 production **RESTful API endpoints** enabling real-time code documentation preview in the client-facing web interface.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs | Noida

- Built a modular LLM + robot control codebase for a food-preparation robotic arm, designing clean architectural boundaries that enabled parallel integration with a Computer Vision team working on real-time calibration.
- Integrated Computer Vision capabilities into robotic control loop, optimizing calibration routines and improving robotic arm positioning accuracy for complex food preparation tasks.

Selected Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud Speech-to-Text and Text-to-Speech, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.

- Designed **within-subjects experimental study** ($N = 18$) to evaluate three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) on user perception using Godspeed questionnaire and custom metrics.
- Demonstrated through **Wilcoxon signed-rank test** ($p < 0.05$) that embodied gestures significantly improved perceived animacy and immersion compared to explicit loading indicators.
- **Co-authored paper submitted to MIT's *Presence* journal** (under review), analyzing latency feedback mechanisms in embodied AI agents.

AI Resume Builder & Interview Prep Tool | *Next.js, Tailwind, Firebase, Gemini API* Jan. 2025 – Apr. 2025

- Built full-stack AI tool with real-time PDF preview and ATS-optimized resume generation, implementing **Firestore Authentication** with Google OAuth for secure user management across mobile and desktop.
- Integrated **Gemini 2.5 Flash** to power an “Interview GPT” feature that generates personalized interview questions by cross-referencing the user’s resume profile with a companion interview prep database (DSA problems, STAR templates) – implementing error handling for incomplete or ambiguous resume inputs to ensure relevant question generation.

Kelp Forest Semantic Segmentation | *PyTorch, U-Net* Jan. 2024 – May 2024

- Achieved **Top 100 ranking (83rd percentile)** in a DrivenData satellite imagery competition using **U-Net with EfficientNet-B3** backbone for semantic segmentation of kelp forests in noisy real-world satellite imagery.
- Built streak-detection pipeline using an ensemble of **CNN classifiers** (ResNet, DenseNet) with hard voting to identify and filter atmospherically corrupted satellite images from training data, achieving **F1 score of 0.952** on streak classification.

Capacitated Vehicle Routing with Time Windows | *Google OR-Tools, Google Maps API* Sept. 2023 – Dec. 2023

- Modelled a **multi-constraint VRP** end-to-end – formulating capacity, time-window, and depot constraints; tuning OR-Tools solver search strategies; and designing the full data pipeline from raw inputs to optimized routes.
- Integrated **Google Distance Matrix API** for real-world transit data and built interactive Maps visualizations for route validation and multi-stop delivery planning.

Leadership & Involvement

Head of Technology Apr. 2025 – Present

Athleda – Plaksha Sports Society | *Plaksha University*

- Personally design and build a **sports platform serving 500+ students** alongside 2 junior developers – shipping event registration, photo galleries, and a live-scoring sports fest site with real-time score updates.
- Mentor junior developers through code reviews and pair programming, maintaining production quality across an actively growing platform.

LEAP.AI @ Plaksha Feb. 2024 – Feb. 2025

Web Developer → *Head of Web Development* | *Plaksha University*

- Contributed as a core developer on a team of 7–8 building the AI club website, then led the team through **2 full redesigns** as Head, growing the platform to **800+ monthly users** for events, workshops, and project showcases.